



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
UNIT TEST
05-12-2021

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

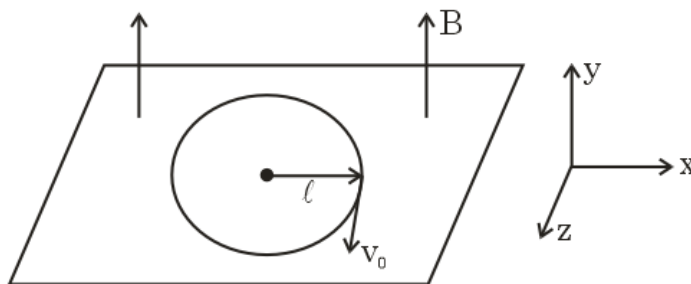
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A tunnel is made along the diameter of earth. Assume that earth is having a hard core till a radius of 3200 km and from a radius of 3200 km to 6400 km (surface of the earth) acceleration due to gravity is constant at 10 m/s^2 . Inside the hard core, the acceleration increases linearly from zero at center to 10 m/s^2 at radius of 3200 km.
 - (A) This implies that the mass density from 3200 km to 6400 km is zero.
 - (B) This implies that the mass density from 3200 km to 6400 km is varying inversely as distance from the center.
 - (C) This implies that the mass density from center to 3200 km is constant.
 - (D) This implies that the mass density from center to 3200 km increases linearly with distance from the center.

Space for Rough Work

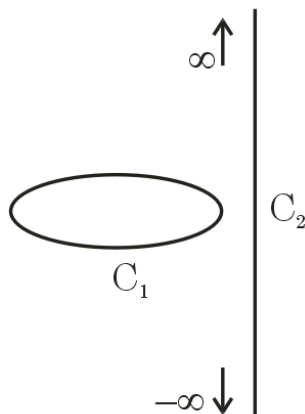
2. A small dipole $\vec{P}$ is placed inside an irregular cavity of a spherical conductor having a charge Q on it. Choose the incorrect option(s)
- (A) Net force on dipole is zero
- (B) The charge on outer surface of conductor is non uniform
- (C) The electric field at a distance r from the dipole in its line outside conductor due to charge on outer surface of conductor is $\frac{2KP}{r^3}$.
- (D) Force on sphere is zero
3. An elastic string of force constant k and negligible length is attached to a point on a smooth horizontal table. A particle of charge Q , mass m is attached to the other end. A vertical magnetic field B is present. The particle is pulled out so that length of string becomes ℓ . It is now projected with a velocity v_0 so that the particle moves in a circular path about the other end. What is possible value of $\vec{v}_0$? Take $\ell = 1\text{m}$, $m = 0.2\text{kg}$, $q = 1\text{C}$, $k = 30\text{ N/m}$, $B = 1\text{T}$.



- (A) $15 \hat{k} \text{ m/s}$ (B) $5 \hat{k} \text{ m/s}$ (C) $-10 \hat{k} \text{ m/s}$ (D) $-20 \hat{k} \text{ m/s}$

Space for Rough Work

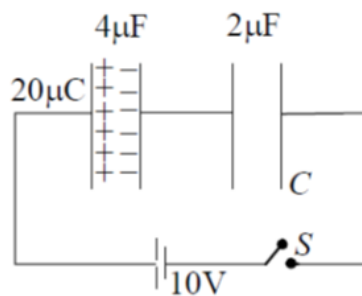
4. A circular coil C_1 and a long straight wire C_2 are kept such that straight wire is in a plane $\perp$ to that of the circular coil. Choose the correct statement(s) :-



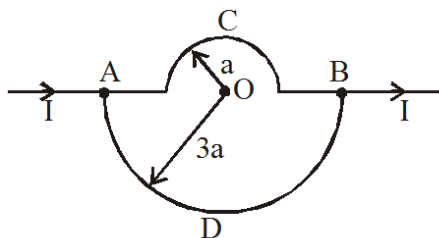
- (A) If a steady current flows in C_2 , $\oint \vec{B} \cdot d\vec{\ell}$ over C_1 is zero.
- (B) If a steady current flows in C_1 , $\int_{-\infty}^{\infty} \vec{B} \cdot d\vec{\ell}$ over C_2 is zero.
- (C) If a steady current flows in C_1 , there are two points on C_2 where $\vec{B}$ is perpendicular to C_2 .
- (D) If a steady current flows in C_1 , $\int_{-\infty}^{\infty} \vec{B} \cdot d\vec{\ell}$ over C_2 is non zero.

Space for Rough Work

5. A $4\mu\text{F}$ capacitor is given $20\mu\text{C}$ charge and is connected with an uncharged capacitor of capacitance $2\mu\text{F}$ as shown in figure. When switch S is closed.



- (A) Charge flown through the battery is $\frac{40}{3}\mu\text{C}$ (B) Charge flown through the battery is $\frac{20}{3}\mu\text{C}$
 (C) Work done by the battery is $\frac{200}{3}\mu\text{J}$ (D) Work done by the battery is $\frac{100}{3}\mu\text{J}$
6. In the given diagram, there are two semicircular parts one having radius 'a' and another having radius '3a' as shown. If resistance of ACB part is R and resistance of ADB part is $3R$. Select the correct option(s) :-



- (A) Current through ACB is $I/4$
 (B) Current through ADB is $I/4$
 (C) Magnitude of $\vec{B}$ at 'O' due to ACB part is $\frac{3\mu_0 I}{16a}$
 (D) Magnitude of $\vec{B}$ at 'O' due to ADB part is $\frac{\mu_0 I}{48a}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

At the vertices of an equilateral triangle with side d , there are three small charged bodies. One of the bodies is fixed, the other two have the mass of M each and charge Q and are free.

7. What charge do you need placed on the fixed body so that when released the other two have minimum acceleration?

(A) $-\frac{Q}{2}$

(B) $-\frac{Q\sqrt{3}}{2}$

(C) $-\frac{Q}{\sqrt{2}}$

(D) $-\frac{Q}{3}$

8. What will be the magnitude of this acceleration?

(A) $\frac{kQ^2\sqrt{3}}{2Md^2}$

(B) $\frac{kQ^2}{2Md^2}$

(C) $\frac{kQ^2}{\sqrt{3}Md^2}$

(D) $\frac{kQ^2}{\sqrt{2}Md^2}$

Space for Rough Work

Paragraph for Questions 9 and 10

We Connected an ideal battery to two identical voltmeters connected in series, and in parallel to one of them we connected a milliammeter. Voltmeters show 1.5 V and 0.3 V, milliammeter shows 0.5 mA.

9. What will the devices show if they are connected to the batteries by connecting in series?

(A) 0.33 mA, 0.8 V each

(B) 0.25 mA, 0.7 V each

(C) 0.45 mA, 0.6 V each

(D) 0.25 mA, 0.8 V each

10. What can be their reading, if all devices are connected to the battery in parallel?

(A) 3 mA, 1.5 V each

(B) 2 mA, 1.7 V each

(C) 3 mA, 1.8 V each

(D) 1 mA, 1.8 V each

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

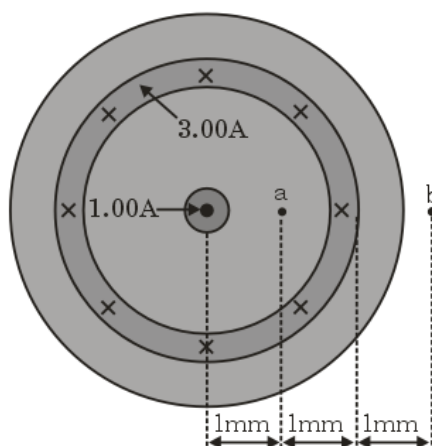
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Figure is a cross-sectional view of a coaxial cable. The center conductor is surrounded by a rubber layer, which is surrounded by an outer conductor, which is surrounded by another rubber layer. In a particular application, the current in the inner conductor is 1.00 A out of the page and the current in the outer conductor is 3.00 A into the page.



1. What is the magnetic field at a (in G)?
2. What is the magnetic field at b (in G)?

Space for Rough Work

Paragraph for Questions 3 and 4

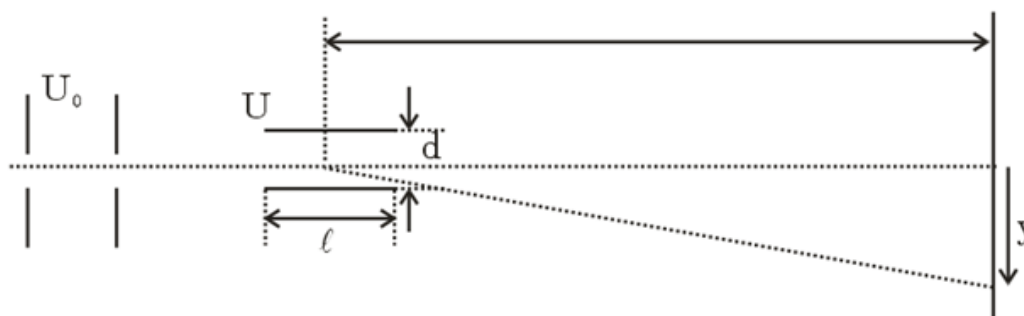
An inclined plane of angle of elevation α is placed to a horizontal surface, and then an object of mass m is placed to the top of the inclined plane. The small mass can slide on the slope without friction. With the help of the horizontal plane it is assured that the slope accelerates in the vertically upward direction at an acceleration of $a < g$.

Data: $\alpha = 30^\circ$, $L = 1.5$ m, $m = 4$ kg, $a = 6$ m/s².

3. How long (in sec) does it take for the object to reach the bottom of the slope, if the length of the slope is L ?
4. What is the force (in N) exerted by the object on the slope?

Paragraph for Questions 5 and 6

A cathode ray oscilloscope is shown here. The beam of electron accelerated by a potential difference of U_0 is made to pass through a capacitor as shown. The vertical down deflection y is used to estimate the potential difference U across the capacitor. Data: the length of the deflecting plates is $\ell = 2$ cm, their distance is $d = 0.5$ cm, the distance between the screen and the center of the plates is $s = 20$ cm, the accelerating voltage is $U_0 = 1000$ V, and the greatest value of the deflecting voltage is $U_{\max} = 100$ V. (Neglect gravity)



5. What is the approximate angle of deflection (in rad) of the electron beam if $U = 50$ V?
6. Determine the sensitivity of the cathode-ray oscilloscope shown in the figure in the unit of mm/volt.

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. A long thick cylinder of radius R (uniformly charged with density ρ coulomb/m³) has a spherical cavity of diameter R touching its axis. If electric field at centre of cavity is $\frac{\rho R}{\eta \epsilon_0}$, find n .
 8. Anant and Balaram are driving their cars along the same straight road in same direction with the constant speed of 33 mph and 63 mph, respectively. Their friend Chaitanya is driving along a different straight road at the constant speed of 52 mph. (All these velocities are measured relative to Earth.) During the motion, the distance between Chaitanya and Anant always remains equal to that between Chaitanya and Balaram. Find Chaitanya speed (in mph) relative to Balaram.
 9. The distance between the parallel plates of an isolated charged condenser is $d = 5\text{cm}$ and the intensity of the field $E = 300 \text{ V cm}^{-1}$. A slab of dielectric constant $K = 5$ and 1 cm wide is inserted parallel to the plates. Determine the potential difference between the plates, after the slab is inserted, (in volt)
-

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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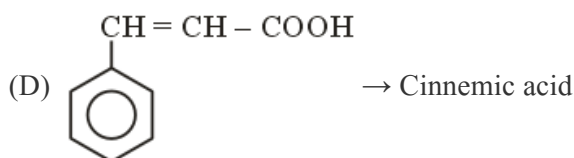
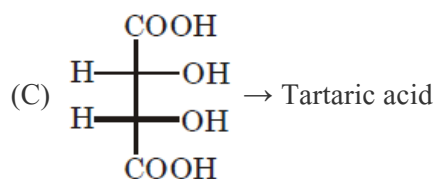
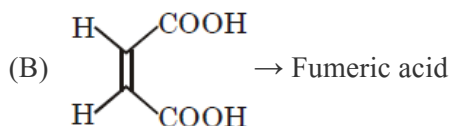
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

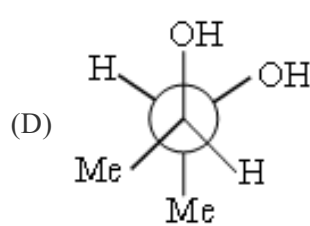
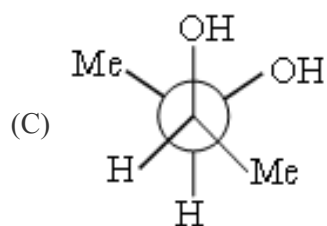
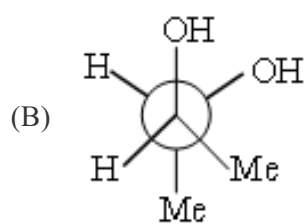
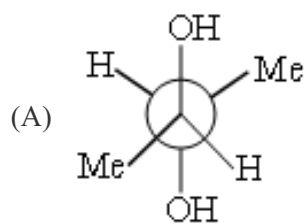
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Identify compounds correctly matched with their common name?



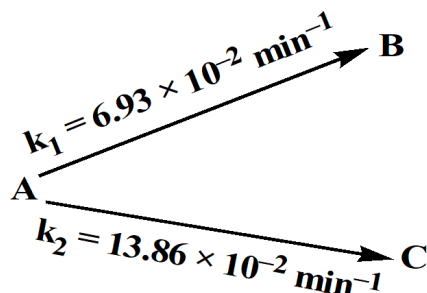
Space for Rough Work

2. Among the following, the conformation that corresponds to the most stable conformation of *meso*-butane-2,3-diol is –



Space for Rough Work

3. Consider the reaction.

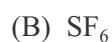


A, B and C all are optically active compound. If optical rotation per unit concentration of A, B and C are 60° , -72° and 42° respectively and initial concentration of A is 2 M, then select correct statement(s).

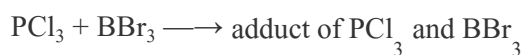
- (A) Solution will be optically active and dextrorotatory after very long time
 (B) Solution will be optically active and levorotatory after very long time
 (C) Half life of reaction is 15 min
 (D) After 75% conversion of A into B and C angle of rotation of solution will be 36° .
4. An ideal monoatomic gas is expanded from initial state (P_1, V_1, T_1) to final volume $2V_1$, separately using two different process $PV^{3/2} = \text{const} (K_1)$ and $PV^2 = \text{const} (K_2)$ than the correct statements are -
- (A) Magnitude of internal energy change by the gas in second process is greater than first process
 (B) Final pressure of gas in second process is more than first process
 (C) Heat is absorbed in both first and second process
 (D) Magnitude of work done in first process is greater than second process

Space for Rough Work

5. Select species in which total lone pair are even in number?



6. Process-(1)



Product(1)

Process-(2)



Product(2)

Select **INCORRECT** statement(s) considering process-1 and 2.

(A) In both process no atom undergoes in rehybridization

(B) No atom is forming bond in excited state in given products

(C) In both process phosphorus atom donate lone pair

(D) In product-(2) total 4 tetrahedral units present

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
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Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

In a H-like species there are two energy levels A and B above the ground state having principal quantum numbers of n_1 and n_2 respectively. A sample of this H-like species has all atoms/ions in excited levels A or B only and none in any other energy level. Energy of level B is greater than that of level A and a total of 15 different lines are emitted from this sample on returning to ground state out of which 6 lines are emitted due to electronic transitions between the levels n_1 and n_2 only. Also energy difference between levels n_2 and n_1 ,

$$E_{n_2} - E_{n_1} = 4.53 \text{ eV.}$$

7. The correct option is -

(A) $n_2 = 2, n_1 = 5$ (B) $n_2 = 6, n_1 = 3$ (C) $n_2 = 5, n_1 = 2$ (D) $n_2 = 3, n_1 = 6$

8. Calculate minimum wavelength among all transitions.

(A) $\frac{4}{3R_H}$ (B) $\frac{9}{8R_H}$ (C) $\frac{900}{11R_H}$ (D) $\frac{9}{35R_H}$

Space for Rough Work

Paragraph for Questions 9 and 10

The known sulfur fluorides are quite different from the other halides of sulfur in their stability, reactivity and to some extent even in their stoichiometries ; it is therefore convenient to consider them separately. Moreover, they have proved a rich field for both structural and theoretical studies since they form an unusually extensive and graded series of covalent molecular compounds in which S has the oxidation states 1, 2, 3, 4, 5 and 6 and in which it also exhibits all coordination numbers from 1 to 6 (if SF_5^- is also included). The compounds feature a rare example of structural isomerism amongst simple molecular inorganic compounds (FSSF and SSF_2) and also a monomer-dimer pair (SF_2 and F_3SSF).

9. Two structural isomers are possible for sulfur fluoride with S_2F_2 molecular formula. Select **CORRECT** statement :-
- | | |
|---------------------------------------|--|
| (A) Both isomers are polar and planar | (B) Same S-S bond length in both isomers |
| (C) Both isomers are polar | (D) Different number of S-F bonds |
10. Select **CORRECT** statement regarding S_2F_{10} ?
- | | |
|---|---|
| (A) Two octahedra joined by common vertex | (B) Two octahedra joined by common face |
| (C) Two octahedra joined by common edge | (D) None of these |

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

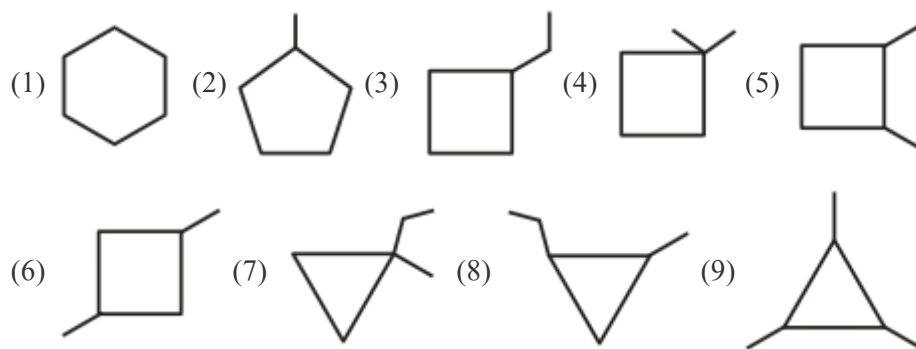
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- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Some possible cyclic structural isomers with molecular formula C_6H_{12} are :



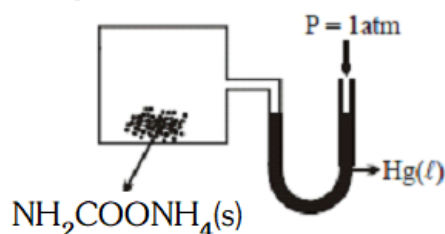
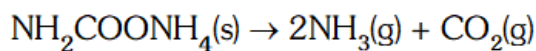
Answer the following questions based on above cyclic isomers

1. How many of these structures are capable of showing optical isomerism
2. Identify total number of structural isomers which can show geometrical isomerism

Space for Rough Work

Paragraph for Questions 3 and 4

A manometer attached to 36 litre flask contains some inert gas & ammonium carbamate $[\text{NH}_2\text{COONH}_4(\text{s})]$ having no difference in mercury level initially as shown. $\text{NH}_2\text{COONH}_4(\text{s})$ decomposed completely on heating according to reaction



If same amount of $\text{NH}_2\text{COONH}_4(\text{s})$ is decomposed completely in separate container and produced $\text{NH}_3(\text{g})$, required 500 ml of $\frac{2M}{3} \text{H}_3\text{PO}_4$ solution for complete neutralisation. During the process temperature remains at 300K. $[R = 0.08 \text{ atm-litre mole}^{-1}\text{K}^{-1}, 1 \text{ atm} = 76 \text{ cm of Hg}]$

3. The difference in height of Hg column (in mm) after complete decomposition of $\text{NH}_2\text{COONH}_4(\text{s})$ is:
4. Number of moles of $\text{CO}_2(\text{g})$ produced is

Space for Rough Work

Paragraph for Questions 5 and 6

Complete the following table by matching the species SO_4^{2-} , SO_3^{2-} , S_2O (linked in the non-linear form of S-S-O), and $\text{S}_2\text{O}_7^{2-}$ (linked in the non-linear form $[\text{O}_3\text{S-O-SO}_3]^{2-}$) with the corresponding S-O bond distances in the last column. Note that there are two different S-O distance (terminal and bridging) in $\text{S}_2\text{O}_7^{2-}$.

	Species	S-O bond length (Å)
1		1.645
2		1.51
3		1.49
4		1.437
5		1.43

5. Value of S-O bond length in S_2O from given table is _____ Å.
6. Value of S-O bond length in SO_3^{2-} from given table is _____ Å.

Space for Rough Work

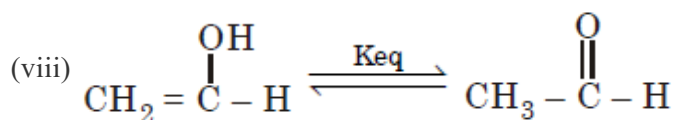
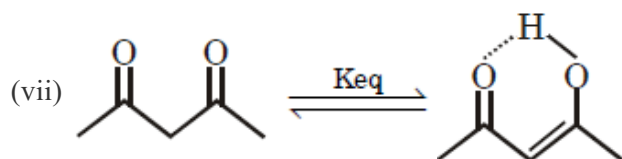
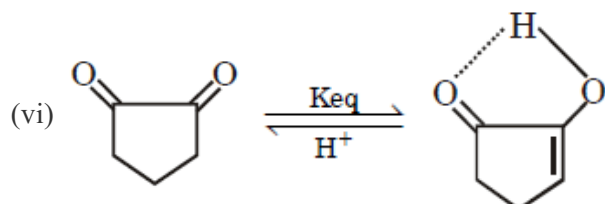
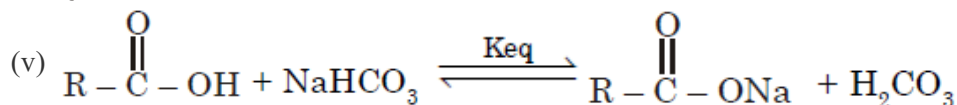
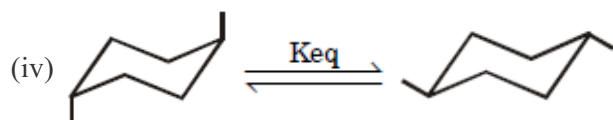
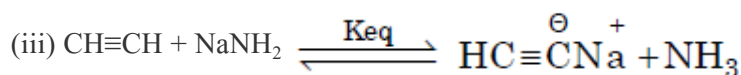
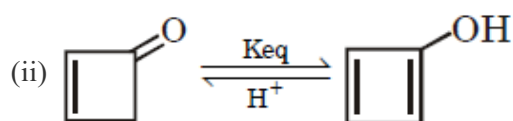
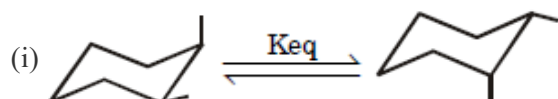
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

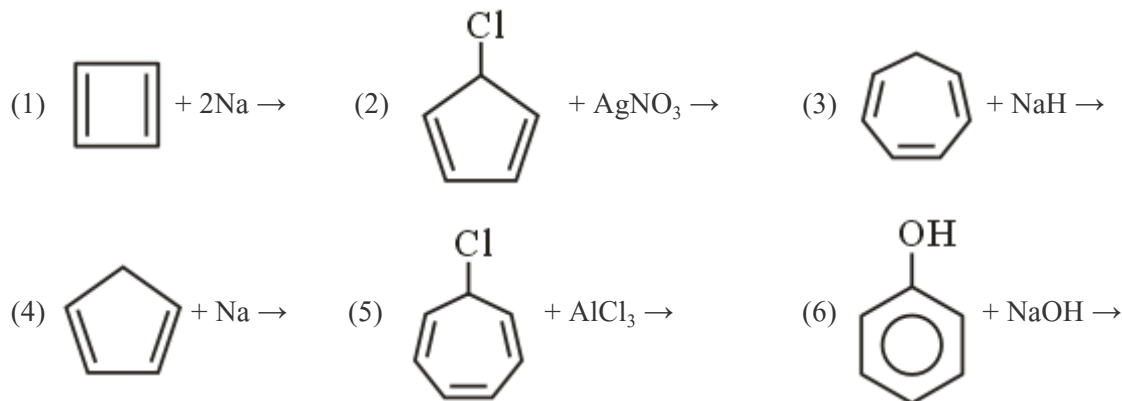
Zero Marks : 0 In all other cases

7. In how many options given below the value of $K_{eq} > 1$

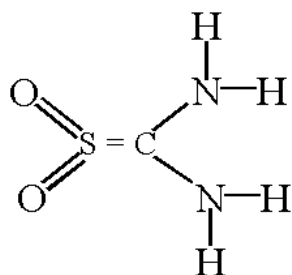


Space for Rough Work

8. How many of the following reaction gives aromatic product-



9. Consider the only given resonating structure of given molecule, calculate value of (x + y).



Here x = Number of d orbitals of S involved in bonding

y = maximum number of atoms that may lie in a plane

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following functions are injective in their respective domain ?

(A) $f(x) = e^{|x|} - e^{-x} + 2$

(B) $f(x) = \frac{2^x + 1}{4^x - 1}$

(C) $f(x) = 4x - |\cos x|$

(D) $f(x) = x^3 - 2x^2 + 5x + 6$

Space for Rough Work

2. From the point with abscissa 25, two tangents are drawn to the ellipse $24x^2 + 25y^2 = 600$ with foci at S_1 and S_2 . The points of contact of tangents are A and B. If the distance of A from S_1 is $\frac{60}{13}$ units, then
- (A) The distance of A from S_2 is $\frac{80}{13}$ units
- (B) The distance of B from S_1 is 5 units
- (C) The distance of B from S_2 is 5 units
- (D) The distance of A from the directrix corresponding to S_2 is $\frac{350}{13}$ units.
3. 8 players compete in a tournament and there is exactly one match between every two players. The winner gets 1 point, loser 0 point and each gets $\frac{1}{2}$ point if the match is drawn. The final result is known where everyone gets different score and player placed second gets the same score as the total of four bottom players, then which of the following statements is/are correct ?
- (A) The total of the player placing second is 6
- (B) The total score of the top four players is 22
- (C) The score of winner is 7
- (D) The total score of third & fourth players is 9.

Space for Rough Work

4. An ellipse having eccentricity $\frac{1}{\sqrt{2}}$ and latus rectum 4 units, has $(3, 2\sqrt{3})$ as one of its focus. If (α, β) is its other focus then
- (A) minimum $\left\{ \sqrt{\alpha^2 + \beta^2} \right\} = 4\sqrt{2} - \sqrt{21}$ (B) maximum $\left\{ \sqrt{\alpha^2 + \beta^2} \right\} = \frac{11}{4\sqrt{2} - \sqrt{21}}$
- (C) $(\alpha - 3)^2 + (\beta - 2\sqrt{3})^2 = 32$ (D) $(\alpha - 2\sqrt{3})^2 + (\beta - 3)^2 = 32$
5. All the 7 digit numbers containing each of the digits 2, 3, 5, 6, 7, 8, 9 exactly once and not divisible by 5 are arranged in increasing order then which of the following is/are true.
- (A) 2017th number in the list is 6325789 (B) 1562th number in the list is 5723698
- (C) 818th number in the list is 3625798 (D) 2123th number in the list is 5236897
6. Circles are drawn with OA & OB as diameters, where A & B are points of parabola $y^2 = 4x$. These circles meet at P (other than O). m_1 and m_2 are slope of tangents to parabola at A & B respectively and m is slope of chord AB, then (given $m_1 + m_2 \neq 0$, A, B are points other than origin and 'O' is origin)
- (A) A, P, B are collinear points (B) m is harmonic mean of m_1 and m_2
- (C) m is arithmetic mean of m_1 and m_2 (D) OP is perpendicular to AB

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let a function $f(x) = \log_e \left(\sum_{r=1}^n \frac{1}{(x+r)(x+r+1)} \right)$.

On the basis of above information, answer the following questions :

7. Number of integers those are not in domain of $f(x)$, if $n = 9$, is -
 (A) 8 (B) 9 (C) 10 (D) 11
8. Number of common positive integers in range of $f(x)$ for $n = 8$ and $n = 10$, is
 (A) 0 (B) 2 (C) 18 (D) infinite

Space for Rough Work

Paragraph for Questions 9 and 10

For $n \in \mathbb{N}$ consider the identity $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$

9. If n is not a multiple of 3 and $\sum_{r=0}^n (-1)^r a_r \binom{n}{r} = k$, then find k .

(A) 0

(B) 1

(C) 2

(D) 3

10. If $a = \sum_{k=1}^{[n/2]} \binom{n}{2k} \binom{2k}{k}$ (where $[.]$ denotes the greatest integer function) then find $(a_n - a)$.

(A) 2

(B) 1

(C) 4

(D) 5

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A Rod of length 10cm slides between two lines, x-axis & $y = -\sqrt{3}x$. If locus of mid point of rod is curve 'S'.

1. Length of intercept made by S on the X-axis is -
2. Maximum distance between origin and a point on the curve S is-

Paragraph for Questions 3 and 4

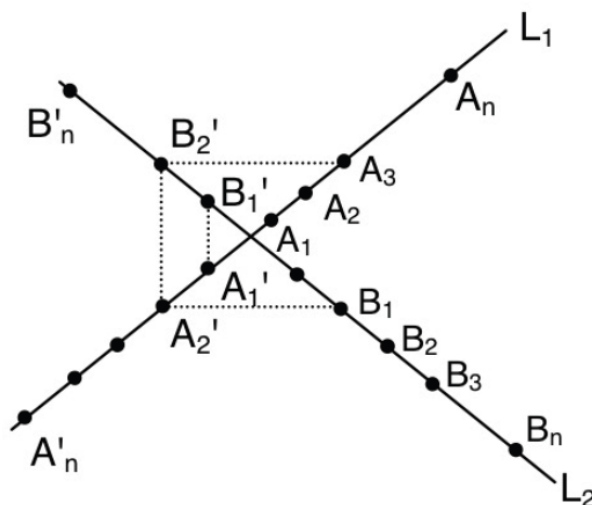
Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a hyperbola such that its eccentricity and eccentricity of its conjugate hyperbola are same. Let foot of the perpendicular drawn from focus of H to its one of the tangent is $\left(\frac{1}{2}, \frac{\sqrt{15}}{2}\right)$. Consider the circle $C : x^2 + y^2 = |a|^3 - 4b^2 + 12$.

3. Number of common tangent of H & C is
4. Let $E : \frac{x^2}{16} + \frac{y^2}{\beta^2} = 1$ be the ellipse which cuts H orthogonally then $|\beta|$ is

Space for Rough Work

Paragraph for Questions 5 and 6

Consider the points $A_1, A_2, A_3, \dots, A_n, A'_1, A'_2, A'_3, \dots, A'_n$ on line L_1 and $B_1, B_2, B_3, \dots, B_n, B'_1, B'_2, B'_3, \dots, B'_n$ on line L_2 as shown in figure



5. If number of triangles obtained by joining these points is 1584, then find the value of 'n'
6. If number of quadrilaterals (concave or convex) obtained by joining these points is $\lambda n^4 - \beta n^3 + n^2 + kn + r$, then find the value of $\lambda + \beta$.

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. If the number of solutions of $|3[x] - 4x| = 4$ is λ , then value of λ is (where $[.]$ is greatest integer function)
 8. The remainder when $10^{(10)} + 10^{(10^2)} + 10^{(10^3)} + \dots + 10^{(10^{10})}$ is divided by 7, is
 9. The number of ordered pairs of non-empty sets A and B such that
 (i) $A \cup B = \{1, 2, 3, \dots, 12\}$ and (ii) $A \cap B = \phi$
 and (iii) (number of elements of A) $\notin A$ and (iv) (number of elements of B) $\notin B$
 is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work